TO DO:

- Sam and Loic to combine their markdowns with the main analysis
- Everyone - clean up repo
- Pick visualisations for power point
- Start drafting the power point

Regroup - Thursday AM (Yasmin to send invite). Get script fully done.

By Friday AM: have script and slides ready

PLAN (remove later)

Central Narrative + Section 1: Motivation (2 min)

Speaker notes of ~200 words take ~1.5 mins

CENTRAL NARRATIVE

- Research question: can we transfer TC forecasting models trained on the data-rich WP to the data-scarce SP, despite the hemisphere flip? – Included
- Hypothesis: intensity prediction should transfer well (shared thermodynamic physics like SST and wind shear) while direction prediction should not (movement patterns governed by hemisphere-dependent Coriolis force) - Included
- Key finding preview: no single architecture excels at both tasks. Spatial models (U-Net+FiLM) capture local wind patterns best for direction, spectral models (U-FNO) capture global thermodynamic patterns best for intensity – Doesn't belong on slide 1
- Structure: EDA builds the transfer hypothesis, method tests it with two model families, results confirm the asymmetry, explainability reveals why – Not sure

SECTION 1: MOTIVATION AND RESEARCH QUESTION

- Data scarcity problem: SP has only 30 storms / 922 timesteps vs WP's 117 storms / 3,621 timesteps in TCND. Too few to train reliable deep learning models on SP alone, especially for 8-class direction and 4-class intensity classification - Included
- Transfer learning as the solution: Qu et al. (2025) fine-tuned a WNP model on ENP and outperformed operational forecasting systems, showing cross-basin transfer is viable within the same hemisphere Not sure
- Why WP-to-SP is the hardest and most interesting transfer pair: WP is in the Northern Hemisphere, SP in the Southern Hemisphere, so the Coriolis force reverses. Storms curve NW/W in WP but SE/W in SP. This is the first cross-hemisphere transfer test on TCND - Included
- The hemisphere flip as a natural experiment to separate transferable from non-transferable knowledge: intensity physics (SST drives energy, wind shear inhibits organisation) operates identically in both hemispheres, but storm movement is governed by hemisphere-specific steering flows and Coriolis deflection - Included
- State research question: can we transfer WP-trained models to SP, and does intensity prediction transfer better than direction prediction as the shared-physics argument would suggest? - Included

PLAN (remove later)

Section 2: EDA and Transfer Hypothesis (1.5 min)

- WP and SP are the most statistically similar cross-hemisphere basin pair, with the lowest average Wasserstein distance across physical variables (SST, wind shear, pressure, wind speed). This makes them the strongest candidate for cross-hemisphere transfer and the fairest test of our hypothesis – Not needed, explained qualitatively why WP & SP is a good test pair
- Direction distributions are near-mirror images across hemispheres: WP storms move predominantly NW/W, SP storms predominantly SE/W/SW. A WP-trained direction classifier will systematically predict the wrong quadrant for SP storms unless corrected
- Intensity-related variables (SST, vertical wind shear) have substantially overlapping distributions between WP and SP. The thermodynamic environment that drives intensification looks similar in both basins. DeMaria and Kaplan (1994) identified SST and shear as the two strongest intensity predictors – Included
- Three data modalities, introduced through the lens of what transfers and what doesn't:

Data3D (15-channel spatial grids, 81x81 pixels): u/v wind at 4 pressure levels (200, 500, 850, 925 hPa), SST, geopotential height, plus two derived channels (wind shear magnitude and 850 hPa vorticity computed during preprocessing). Wind U/V components reverse sign across hemispheres, but SST and shear magnitude patterns are basin-agnostic

Env-Data (40 dimensions after cleaning): originally 94 features, but we removed 54 that directly leak basin identity (area one-hot encodings, coordinate bins, basin flags). Remaining features: wind components, movement velocity, intensity class, direction history, shear magnitude, steering flow. Still heavily hemisphere-specific due to direction/movement encoding

Data1D (4 dims): lat/lon offset from storm centre, normalized wind speed, normalized pressure. Minimal track snapshot

Temporal features (6 dims, extracted separately): storm lifecycle progress (0-1), cyclical sin/cos hour-of-day, cyclical sin/cos month-of-year, normalized storm duration. Used to condition models on where the storm is in its lifecycle via FiLM

- Hemisphere-aware preprocessing applied before any modelling: SP direction labels reflected (N/S swapped), meridional wind components flipped in sign, all z-score normalization statistics computed from WP training data only (no data leakage from SP)
- Clear prediction to test: intensity prediction should transfer well because the underlying physics is shared; direction prediction should transfer poorly because movement patterns are fundamentally hemisphere-dependent - – Included

PLAN (remove later)

Section 3: Method (3.5 min)

TWO-STAGE EXPERIMENTAL PIPELINE

- Stage 1, Zero-shot: Train each model exclusively on WP data (1,863 train / 518 val timesteps). Apply hemisphere-aware corrections before evaluating on SP test set (354 timesteps, 12 storms). No SP data seen during training. This measures how much knowledge transfers for free based on shared physics alone
- Stage 2, Fine-tuning: Take the best WP-pretrained models and fine-tune on the SP training split (354 timesteps, 12 storms) with reduced learning rate. Evaluate on SP validation (532 timesteps, 18 storms). This measures how much target-basin data is needed to close the transfer gap, and whether WP pretraining gives a head start vs training from scratch on SP

FIVE ARCHITECTURES IN TWO MODEL FAMILIES

Spatial family (local pattern learners, hypothesised to be better for direction):

- **U-Net baseline (1a):** 4-level encoder-decoder with 32, 64, 128, 256, 512 channel progression, 9.8M params. SE (squeeze-and-excitation) channel attention learns which of the 15 input channels matter most. Residual connections and DropPath (stochastic depth) for regularisation. Global average pooling of grid features fused with env + 1D features before two classification heads (direction + intensity)
- **U-Net + FiLM (1b):** Same U-Net backbone + Feature-wise Linear Modulation conditioned on temporal features. 6-dim temporal input embedded to 64-dim, then generates per-level affine scale and shift parameters (gamma, beta) applied at every encoder and decoder level. 10.0M params. FiLM adds +5.4pp intensity accuracy over baseline by encoding storm lifecycle stage. A storm early in its life behaves differently from one at peak intensity

Spectral family (global pattern learners, hypothesised to be better for intensity):

- **FNO baseline (2a):** 4 spectral convolution layers that learn in frequency domain via 2D FFT. Multiply by learnable mode weights in Fourier space, then inverse FFT back. 10.3M params. Motivation: spectral representations are not tied to a specific grid location or hemisphere, so learned patterns might be more basin-agnostic. Problem: severely overfits with only 1,816 training samples (early stops at epoch 18/80, params-per-sample ratio 5,672:1)
- **FNO v2 + FiLM (2b):** Redesigned to fix overfitting. Reduced to 3 spectral layers, only 0.9M params (496:1 ratio). Added 9-pixel reflect padding around the grid to avoid FFT wraparound boundary artifacts. Added FiLM temporal conditioning at each spectral layer. Trains to full 150 epochs without overfitting
- **U-FNO hybrid (2c):** 3-branch gated architecture that combines spectral path (FFT-based), U-Net path (local convolution), and residual path (identity skip) at each layer. Branches are weighted by learned softmax gates. Model learns to allocate 40% spectral, 30% U-Net, 30% residual, revealing that it prefers frequency-domain features. 1.0M params. Best intensity model overall

Also benchmarked: ResNet-152 as a standard CNN baseline for comparison. Included in the model comparison but not a focus of the narrative.

KEY DESIGN INSIGHT

Spatial convolutions see local wind patterns and gradients (what determines storm movement direction); spectral methods see large-scale thermodynamic structure across the full grid (what determines intensification). FiLM temporal conditioning helps intensity more than direction because lifecycle stage encodes intensification likelihood but not directional information.

PLAN (remove later)

Section 4A: Results (2.5 min)

SECTION 4A: RESULTS AND ANALYSIS

WP in-basin performance (validating that models work before testing transfer):
- U-Net+FiLM: best direction model, 62.9% accuracy, 0.465 macro F1 on 8-class direction
- U-FNO: best intensity model, 62.5% accuracy, 0.499 macro F1 on 4-class intensity
- No single model wins both tasks. The architecture-task trade-off is real and consistent across all models including ResNet-152 baseline

Zero-shot cross-basin transfer (WP model applied directly to SP, no fine-tuning):
- Direction transfer gap is large: -23.9pp for U-Net+FiLM (62.9% to 39.0%), -30.7pp for U-FNO (50.8% to 20.2%)
- Intensity transfer gap is consistently smaller: -19.2pp for U-FNO (62.5% to 43.3%), -20.2pp for U-Net+FiLM (56.4% to 36.2%)
- Hypothesis confirmed: intensity transfers better than direction across all architectures

Fine-tuning with SP data (354 timesteps, 12 storms):
- Direction barely recovers: U-Net+FiLM stays flat at 39.0% (no gain), U-FNO gains +8.2pp to 28.3%
- Intensity improves meaningfully: U-FNO reaches 49.6% (+6.3pp over zero-shot), U-Net+FiLM reaches 39.2% (+3.0pp)
- Implication: WP pretraining provides a useful initialisation for intensity, but direction requires either much more SP data or fundamentally different approaches

PLAN (remove later)

Section 4B: Explainability + Overfitting (1 min)

Explainability (why the architecture split exists):
- Modality ablation on U-Net baseline: removing the grid causes 31pp direction accuracy drop (62.5% to 31.5%) but only 12.6pp intensity drop. Removing env causes only 0.6pp direction drop but 12.0pp intensity drop
- SHAP GradientExplainer: wind shear magnitude and SST-related features have highest Shapley values for intensity. Historical direction and movement velocity dominate direction
- Gradient attribution on grid channels: u_wind and v_wind at 850 hPa have highest mean absolute gradients. The low-level wind field is what the spatial models learn to read for direction
- These findings explain the architecture trade-off: U-Net's local convolutions are ideal for reading wind field spatial structure (direction); U-FNO's spectral layers are ideal for capturing basin-wide thermodynamic conditions (intensity)

Overfitting management (important given extreme data scarcity):
- Only 1,816 WP training samples. FNO baseline (10.3M params, 5,672:1 ratio): catastrophic overfitting, early stops at epoch 18/80
- U-FNO (1.0M params, 551:1): moderate overfitting, early stops at epoch 41/150. FNO v2 (0.9M params, 496:1): fully resolved, trains all 150 epochs
- Lesson: in extreme low-data regimes, architecture design matters as much as model expressivity

PLAN (remove later)

Key Takeaways (1.5 min) + Models for future work

SECTION 5: TAKEAWAYS

- Hypothesis confirmed: intensity transfers across hemispheres because SST and wind shear physics are universal; direction does not transfer because Coriolis-driven movement patterns are hemisphere-specific. The gap is consistent across all architectures

- No single architecture excels at both tasks: spatial models (U-Net+FiLM) specialise in direction via local convolutions, spectral models (U-FNO) specialise in intensity via frequency-domain learning. This is a fundamental result, not a tuning artifact

- FiLM temporal conditioning and spectral methods disproportionately help intensity. Storm lifecycle stage predicts intensification likelihood but not heading. Direction remains a local spatial problem

- Practical implication: transfer learning from a data-rich basin is viable for intensity forecasting in data-scarce basins with minimal fine-tuning. Direction forecasting requires either substantially more target-basin data or explicit domain adaptation to handle the hemisphere flip

- Future work: extension to NI/SI basins to test generality, domain adaptation techniques (e.g., adversarial alignment) for direction transfer, ensemble of spatial + spectral models to get best of both worlds

PLAN (remove later)

Marking Scheme Reminders

The four criteria are:

1. Data and model presentation: covered by sections 2 (EDA), 3 (method), and 4 (results). Show you understand the data deeply, not just superficially.
2. Clarity of graphs and description: every figure should have a stated takeaway. Use clear labels, consistent colour schemes, and large enough fonts.
3. Organisation and flow: the research question is the throughline. Each section should logically lead to the next. Transitions between speakers should be smooth.
4. Depth of technical understanding: demonstrated by your method justifications, result interpretations, and ability to handle Q&A. Don't just report numbers, explain why.

Machine Learning for Tackling Climate Change

Cross-Basin Generalisation for Tropical Cyclone Forecasting using TropiCycloneNet

Samuel Zhang, Loïc Bouxirot, Yasmin Akhmedova

ELEC70122 | Imperial College London | March 27, 2026

Motivation: Why Cross-Basin Transfer?

DRAFT

1 billion
people live in areas affected by tropical cyclones[1]

USD 2.5 trillion
in damages caused by tropical cyclones since 2000[1]

Our Objective: to investigate what **cross-basin transfer learning** can achieve for cyclone direction and intensity classification when **data is limited**

Our Approach:

Configure existing **DL architectures** — Train on **data-rich region** — **Zero-shot** directly on **data scarce-region test set**

Train on **data-rich region** — **Fine tune** on a **small data-scarce region set** — Evaluate on **data scarce-region test set**

Why Western Pacific and South Pacific Basins?

DRAFT

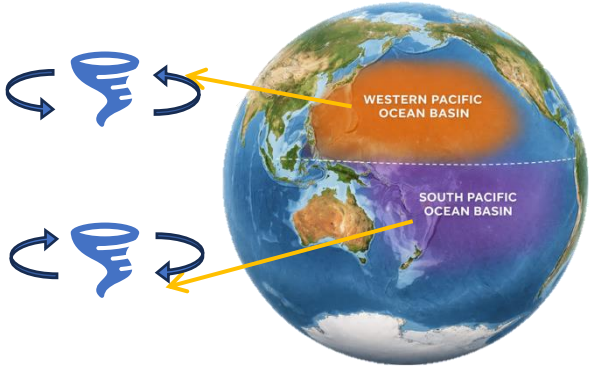
TropiCycloneNet²

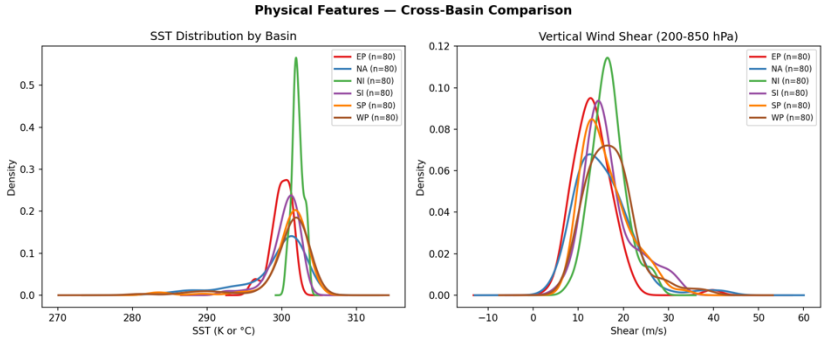
Transfer pair:

- ✓ **South Pacific (SP):** 30 storms, 922 timesteps
- ✓ **Western Pacific (WP):** 131 storms, 4,500+ timestamps

Key classification tasks:

- ✓ **Direction:** storms **curve** in **opposite** directions across hemispheres, so a WP-trained model learns the wrong movement patterns for SP
- ✓ **Intensity:** driven by **sea surface temperature** & **wind shear**, which overlap across both basins[3], so the physics should transfer





Our Hypothesis: intensity prediction should transfer better than direction prediction, because the underlying physics appears to be shared while direction patterns are hemisphere-dependent.

Data & Processing

DRAFT

Data: Spatial snapshot of the **surrounding atmosphere** + Storm's current **position** & **strength** + Summary **environmental conditions**

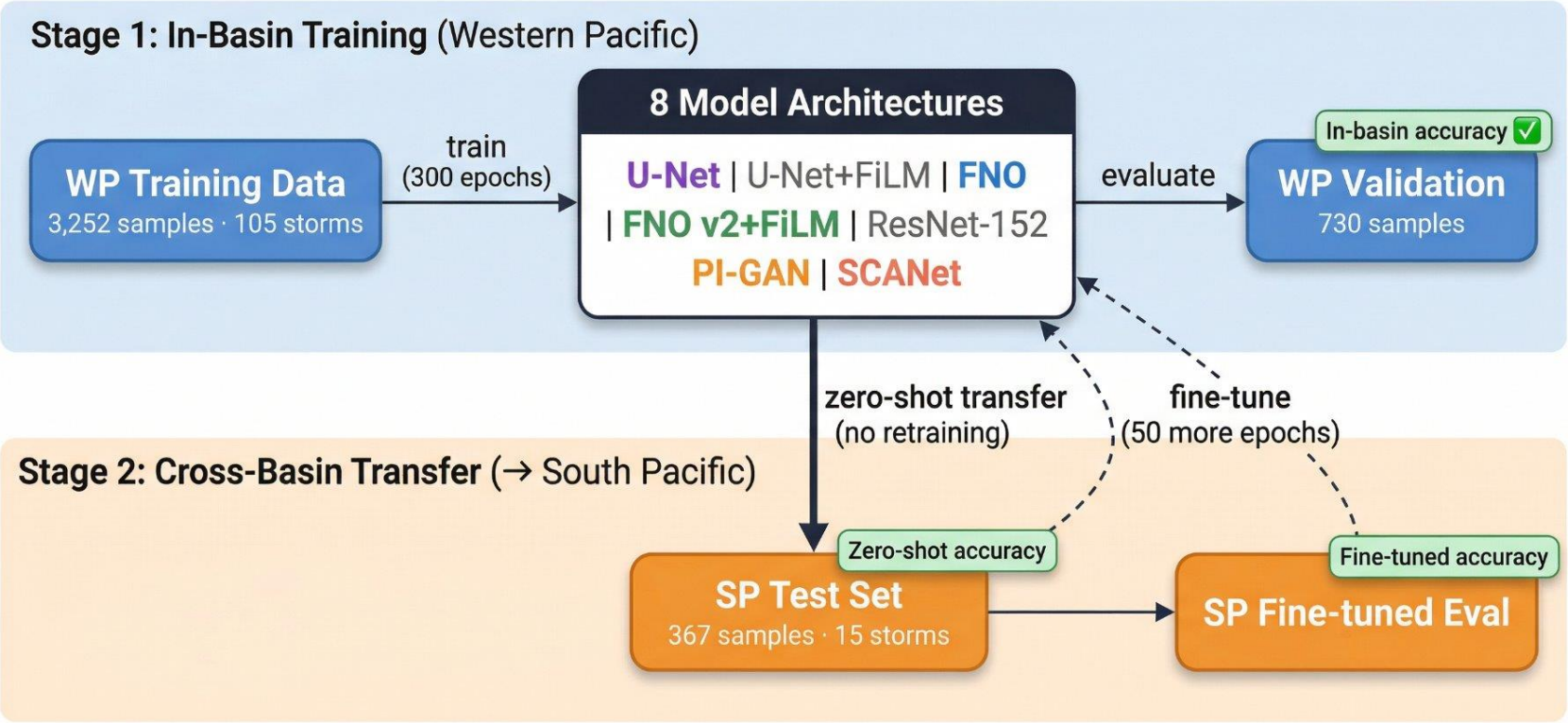
Processing:

- **Basin-leaking removal:** Stripped environmental features encoding basin identity
- **Hemisphere-aware transforms:** Mirrored SP direction labels & flipped the N-S wind component
- **Derived physics features:** Added wind shear & low-level vorticity
- **Normalisation:** All normalisation uses WP training data & during SP inference
- **Storm-level splitting:** Split by entire storms rather than individual timesteps

Method

Two-Stage Pipeline

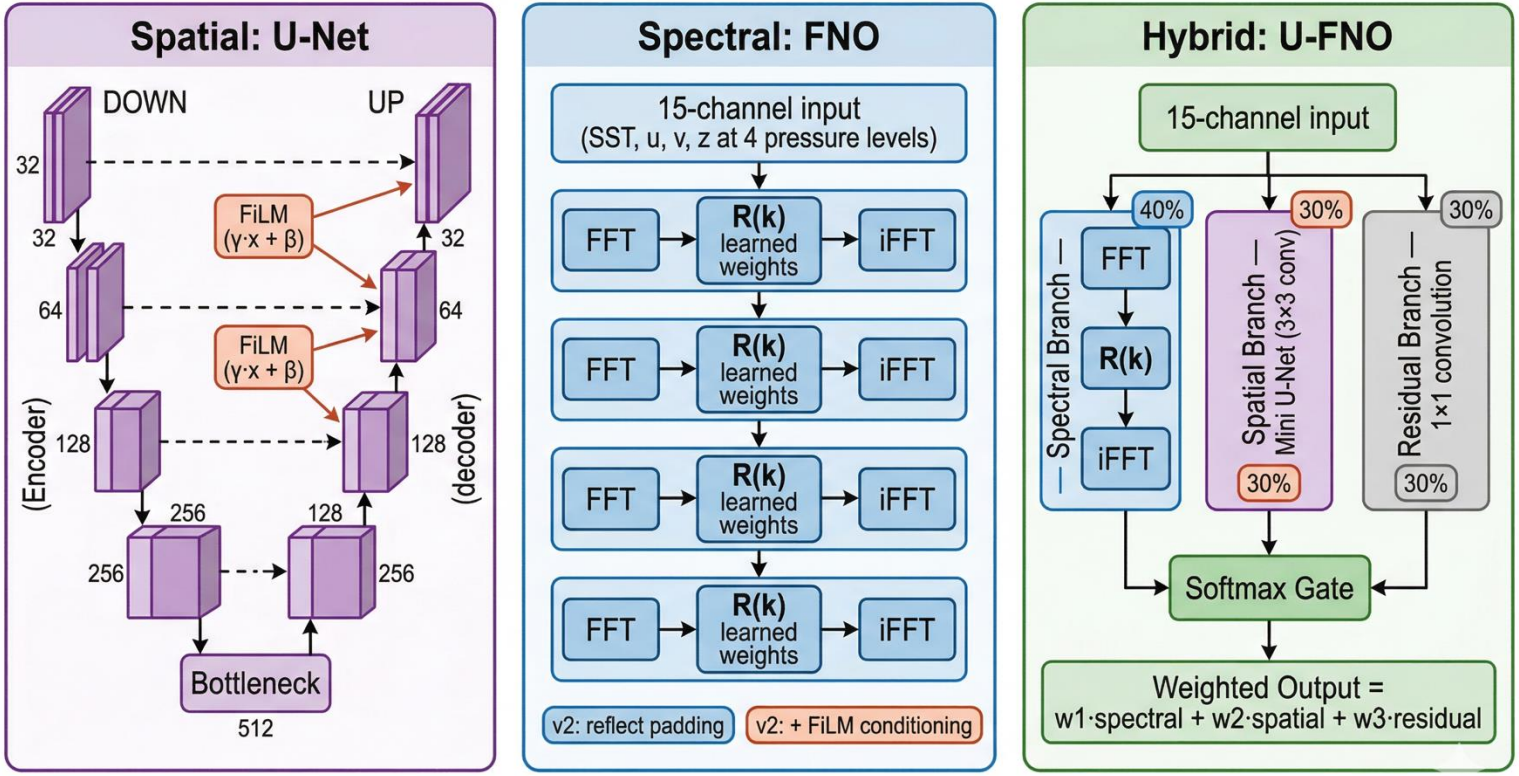
DRAFT



Key question:
Does fine-tuning always help?

Spatial vs Spectral Families

DRAFT



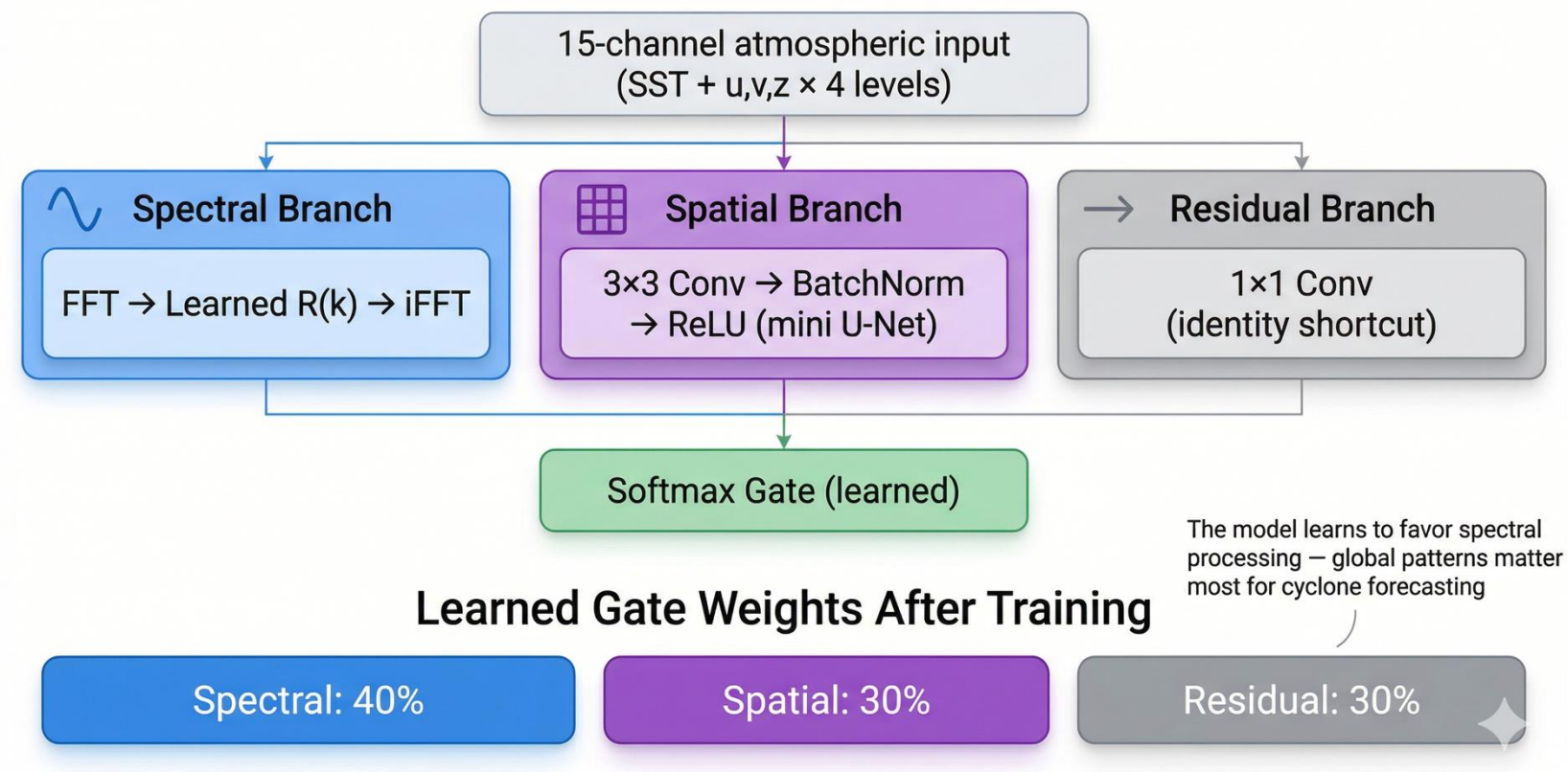
Captures local spatial wind patterns

Global receptive field in every layer

Learns the optimal spectral/spatial mix

U-FNO: Learned Gating

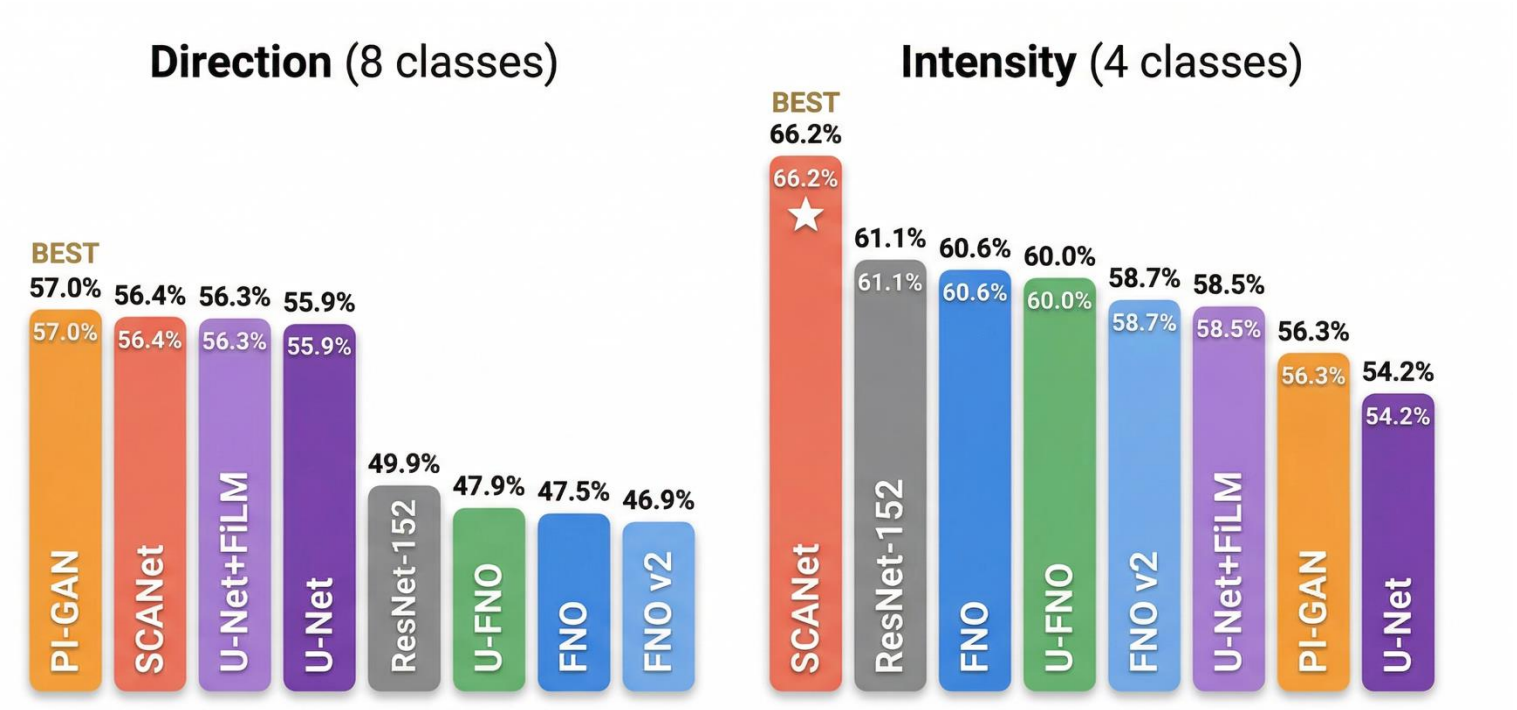
DRAFT



Results and Analysis

WP In-Basin Performance

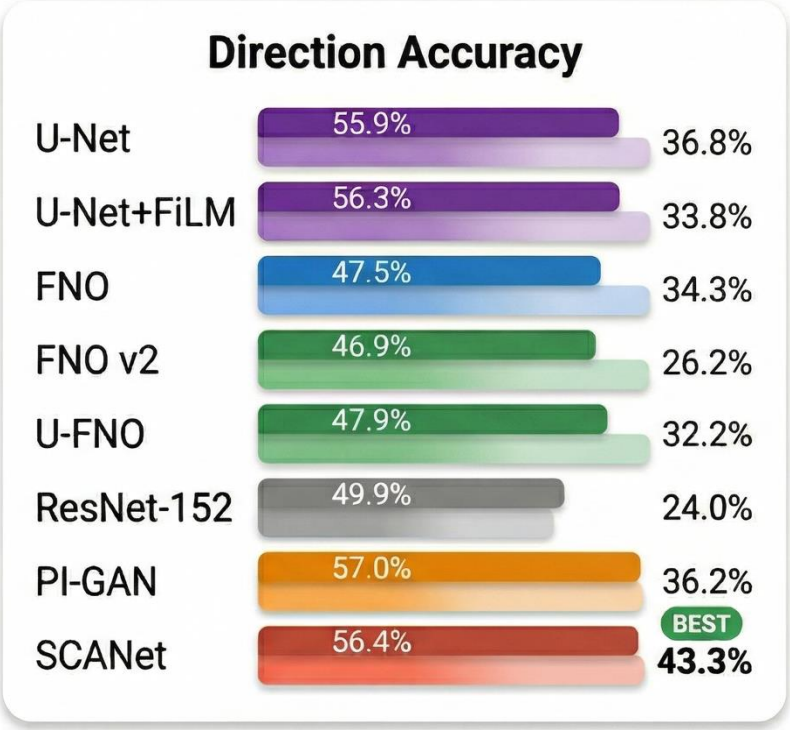
DRAFT

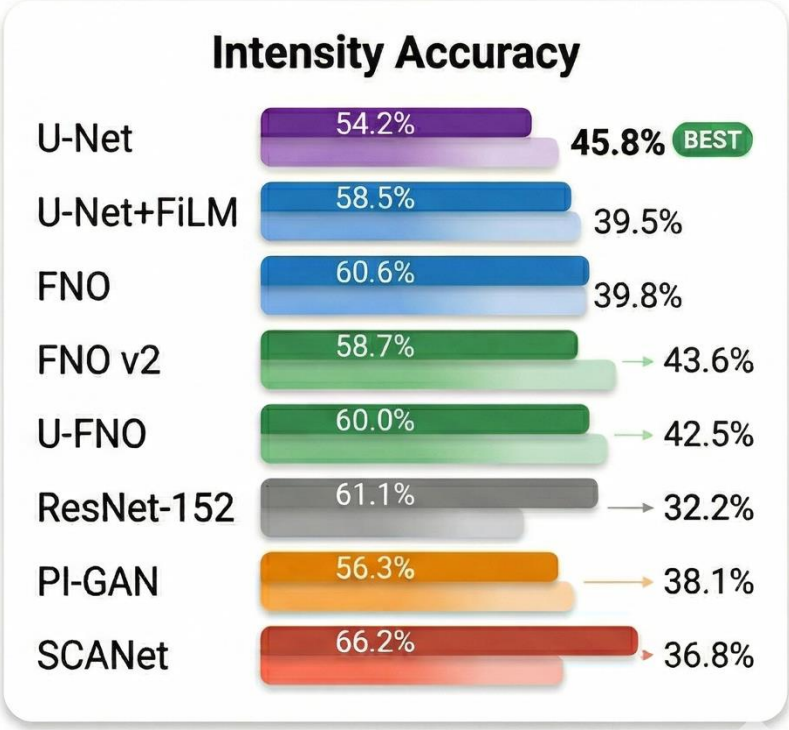


SCANet dominates intensity (+5pp over #2) while PI-GAN leads direction

Zero-Shot Transfer

DRAFT

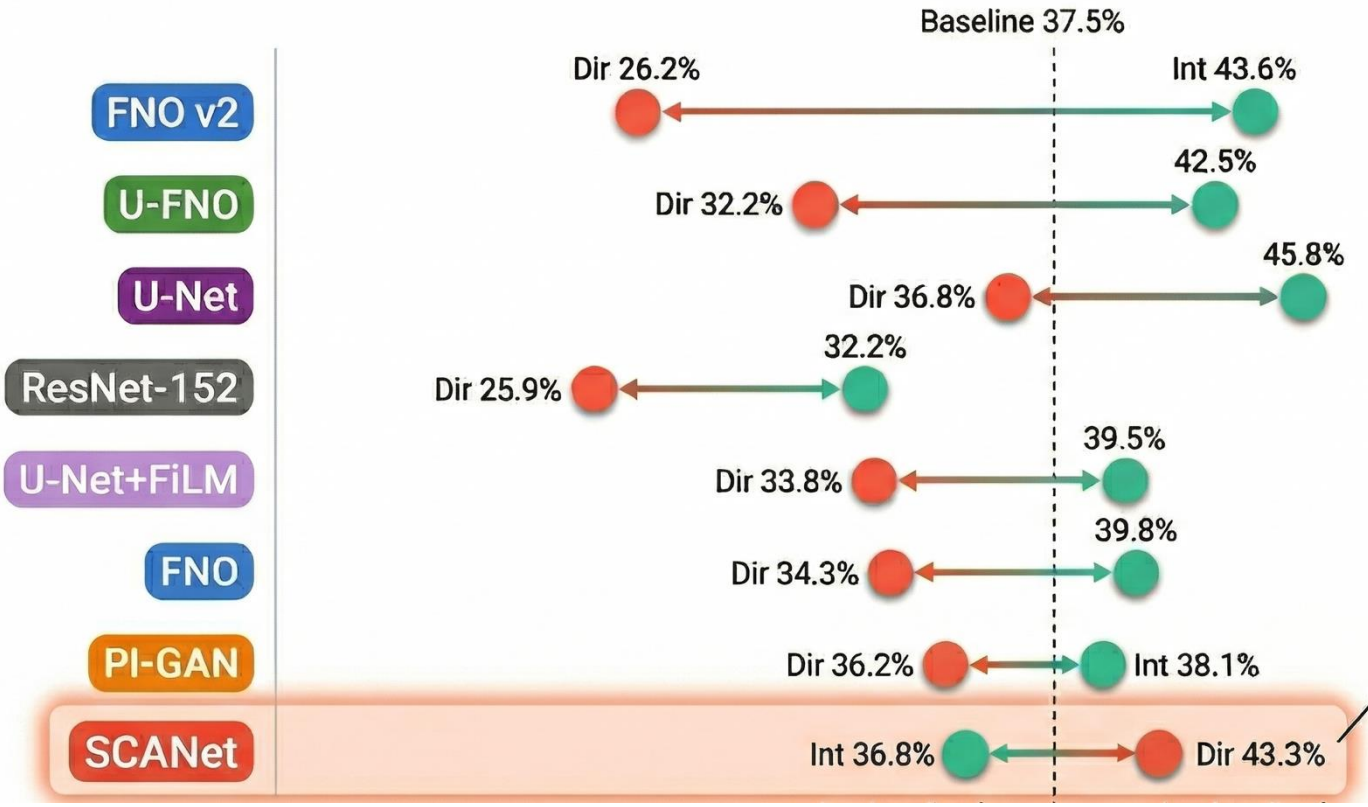




SCANet retains the most direction knowledge; simple U-Net is surprisingly robust for intensity

Direction vs Intensity Transfer Gap

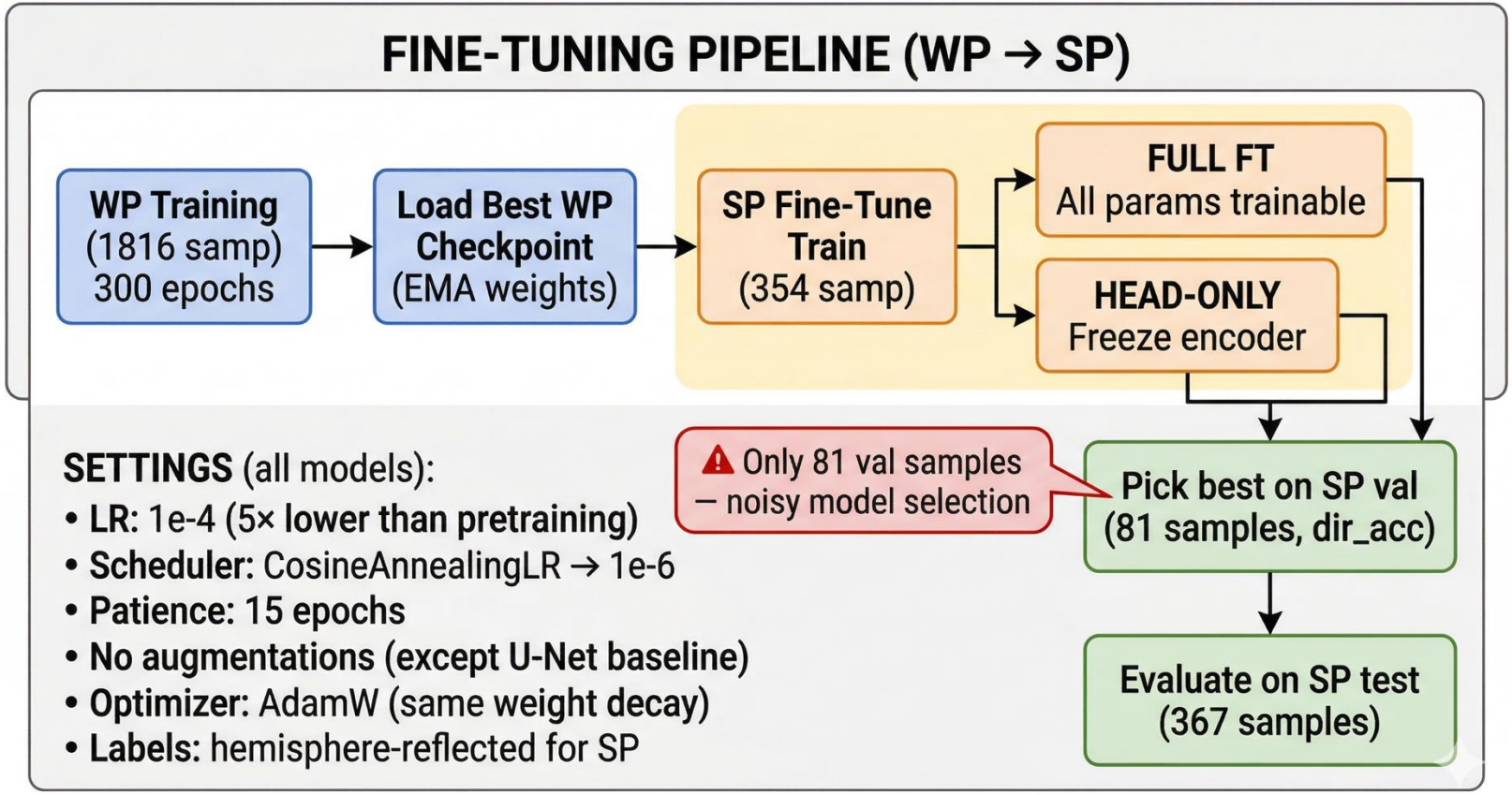
DRAFT



SCANet is the **ONLY** model where direction > **intensity in zero-shot** — its cross-attention captures transferable directional patterns

Fine-Tuning: Double-Edged Sword I

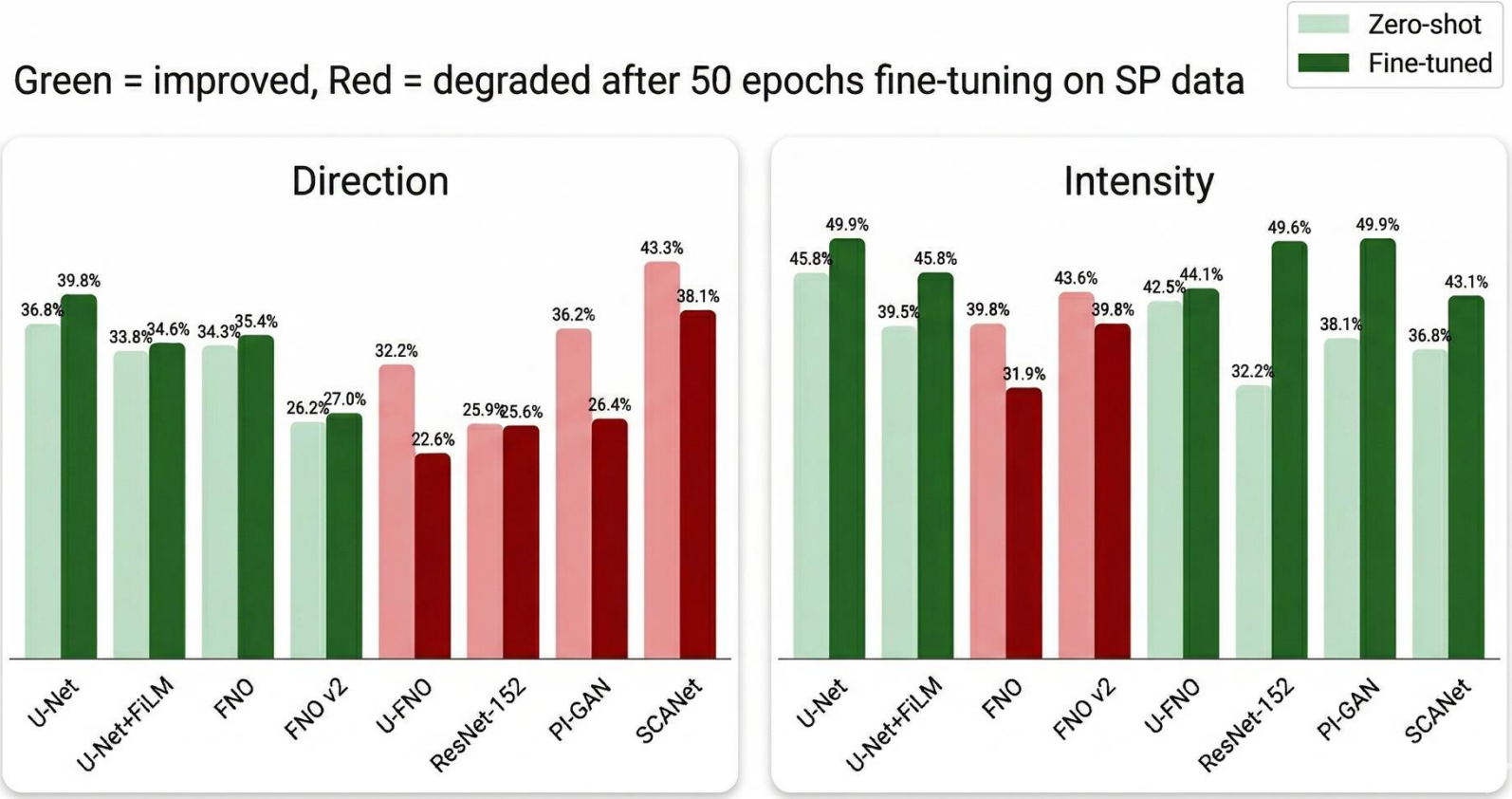
DRAFT



Fine-Tuning: Double-Edged Sword II

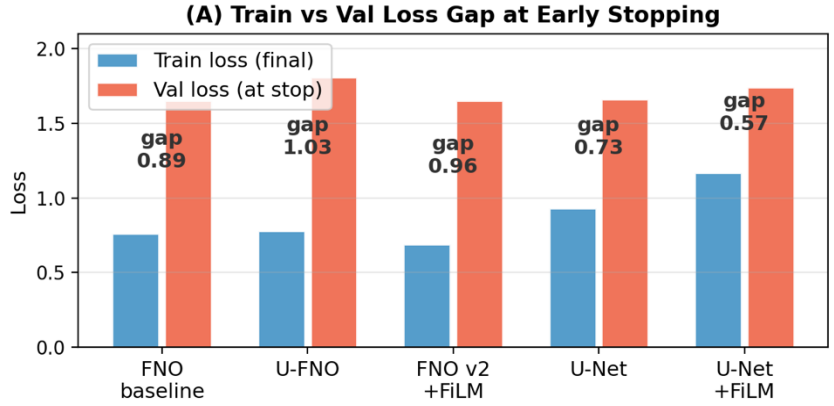
DRAFT

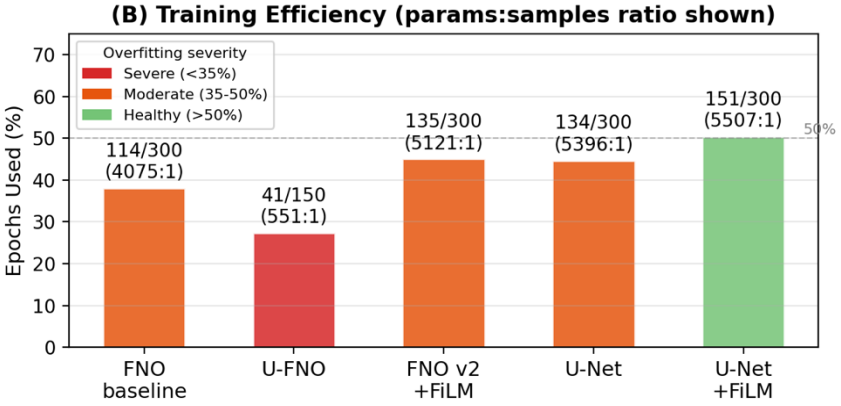
Green = improved, Red = degraded after 50 epochs fine-tuning on SP data

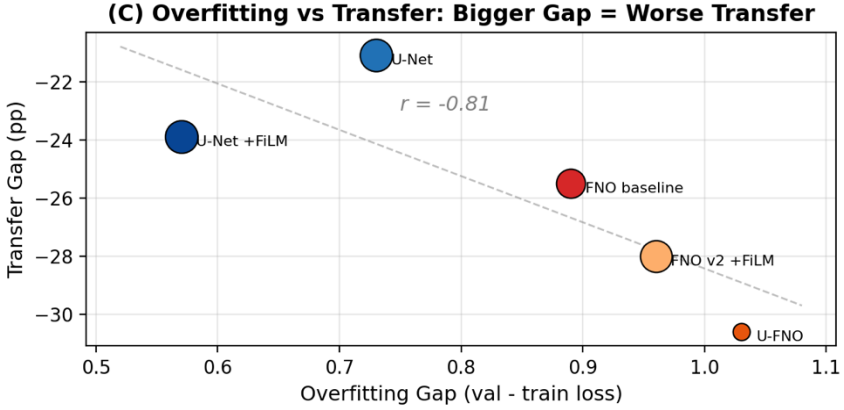


Overfitting Analysis: Data-Limited Regime

DRAFT







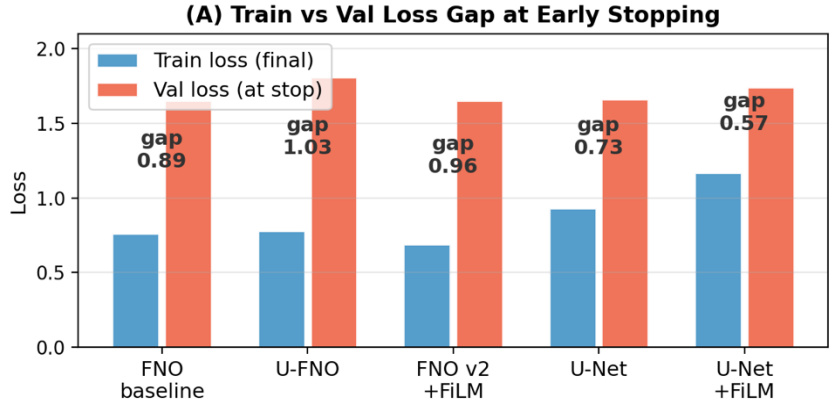
(D) Summary: Overfitting & Generalisation

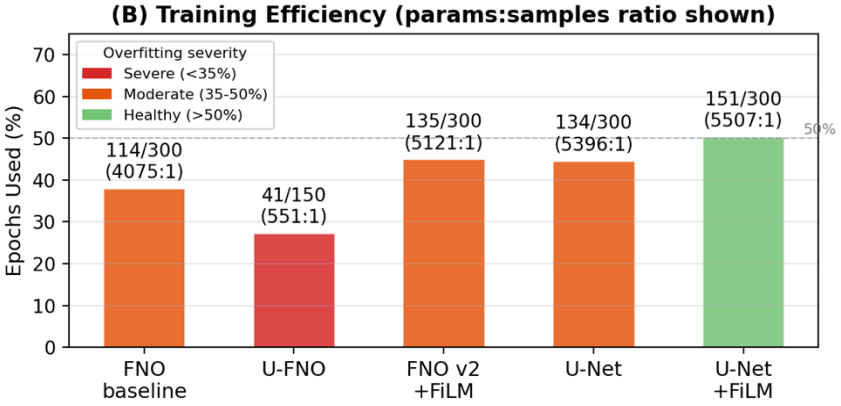
Model	Params	Ratio	Epochs	Overfit Gap	WP Dir	SP Dir
FNO	7.4M	4,074:1	114/300	0.89	50.3%	24.8%
U-FNO	1.0M	551:1	41/150	1.03	50.8%	20.2%
FNO v2	9.3M	5,121:1	135/300	0.96	49.3%	21.3%
U-Net	9.8M	5,396:1	134/300	0.73	62.5%	41.4%
U-Net+FiLM	10.0M	5,507:1	151/300	0.57	62.9%	39.0%

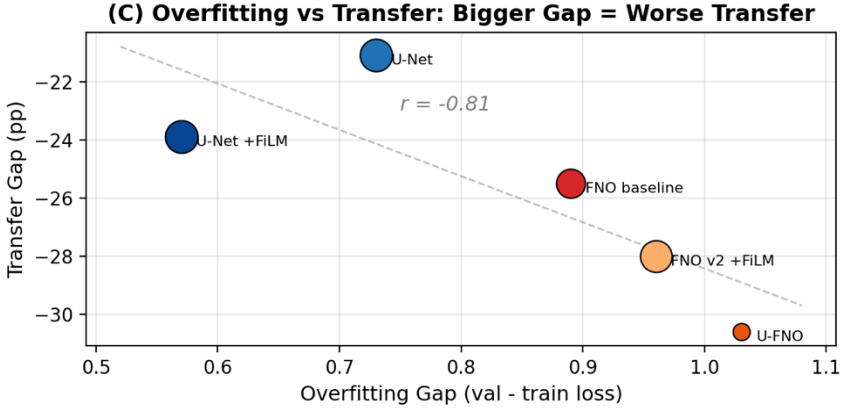
Green = low gap, Red = high. All models trained on 1816 WP samples.

Overfitting Analysis: Data-Limited Regime

DRAFT







(D) Summary: Overfitting & Generalisation

Model	Params	Ratio	Epochs	Overfit Gap	WP Dir	SP Dir
FNO	7.4M	4,074:1	114/300	0.89	50.3%	24.8%
U-FNO	1.0M	551:1	41/150	1.03	50.8%	20.2%
FNO v2	9.3M	5,121:1	135/300	0.96	49.3%	21.3%
U-Net	9.8M	5,396:1	134/300	0.73	62.5%	41.4%
U-Net+FiLM	10.0M	5,507:1	151/300	0.57	62.9%	39.0%

Green = low gap, Red = high. All models trained on 1816 WP samples.

Key Takeaways & Future Work

Slide to be fixed…

✓ **Hypothesis confirmed:** intensity transfers because SST and wind shear physics are universal. Direction does not, because movement patterns are hemisphere-dependent.

✓ **No single architecture excels at both:** spatial models (U-Net+FiLM) specialise in direction via local wind patterns. Spectral models (U-FNO) specialise in intensity via global features.

✓ **Practical implication:** cross-basin transfer is viable for intensity with minimal fine-tuning. Direction requires more target data or explicit domain adaptation.

Future work

1. Test on other basin pairs (NI, SI) to check generality

2. Ensemble spatial + spectral models for best of both worlds – more models like UFNO…

3. Two additional architectures explored:

PI-GAN (physics-informed GAN with auxiliary reconstruction losses)

SCANet (spectral cross-attention with unified context injection)

Thank You!

Questions?

References

[1] Imperial College London, Grantham Institute. "Modelling Tropical Cyclones." Available at: https://www.imperial.ac.uk/grantham/research/climate-science/modelling-tropical-cyclones/

[2] Huang, C. et al. (2025) "Benchmark dataset and deep learning method for global tropical cyclone forecasting." Nature Communications, 16, 5923.

[3] DeMaria, M. & Kaplan, J. (1994) "A Statistical Hurricane Intensity Prediction Scheme (SHIPS)." Weather and Forecasting, 9(2), 209–220.